

CMSC477: Robotics Perception and Planning

Project 3: The Energy-Aware Logistics Challenge

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1. Introduction

In this project, your team will develop an autonomous logistics robot designed to clear a "Loading Dock" by sorting and transporting LEGO bricks to unknown delivery zones. Unlike previous projects, this challenge introduces **resource management constraints**. Your robot must successfully navigate a cluttered environment, locate delivery zones, identify targets (LEGO bricks), and manage a limited battery supply to maximize its delivery score.

2. Environment & Perimeter

- **The Environment:** The **3x3-meter** (10x10-foot) arena is strictly bounded by **orange traffic cones** and **colored tape**.
- **Layout:** The environment is divided into three parts: side 1, obstacle field, and side 2.
 - Side 1 → A recharging station and loading dock.
 - Side 2 → Two goals (one for small bricks and one for large bricks).
 - Obstacle field → A collection of static obstacles dividing the two sides. You can assume there will be no obstacles within Side 1 or Side 2.
- **Boundary Sensing:** Robots can use YOLO or color segmentation to find the boundaries.
- **Loading Dock:** The loading dock will be marked by a colored tape on the ground, and its location will be fixed. The robot can access the dock from any direction.
- **Static Obstacles:** An unknown number of fabric boxes will be placed at random locations within the obstacle field at the start of the session for each team.
- **Delivery Zones (Goals):** There are two goal locations marked by AprilTags. One zone is designated for small bricks and the other for large bricks.
 - **How to Identify:** AprilTags will be placed on one side of a vertical surface (such as a side of a fabric box). You must clearly identify which zone is for small and large bricks. For accuracy, colored tape must be used to **mark a 25x25-cm (~10x10-inch) square** with the AprilTag along one of the sides of the square.
 - **Starting Locations:** The location of each goal will be random.

- **Recharging Station:** The location of the recharging station will be randomized at the beginning of each session.
- **Unknown Locations:** The coordinates for the goals, the loading dock, and the delivery zones are unknown beforehand. The robot must explore and map the arena to find them.

3. Logistics & Scoring Rules

The robot's primary task is to transport LEGO bricks from the loading dock to the correct AprilTag goal without fully depleting its battery (i.e., battery level = 0%).

- **Start Positions:** A referee (TA or instructor) will randomly decide where the robot will start within **side 1** at the beginning of a team's session. This start position will be fixed for a given team. But teams may have different starting positions.
- **LEGO Bricks:**
 - **Placing Bricks at Goals:** To receive points, the robot must place the LEGO bricks **in front of and within 25 cm (10 inches)** of the correct goal's AprilTag.
 - **Knocking over Bricks:** If a LEGO brick is knocked over, no one may stand the brick upright for the remainder of the team's session. However, the robot may continue to move the LEGO brick if it can (e.g., push, pull, or grab the brick).
- **Battery:** The battery drains any time the robot interacts with (e.g., grabs, pushes, or pulls) a brick. If the robot stops and re-engages with a brick, that is counted as one interaction. If the robot interacts with multiple bricks, the battery **depletes by some multiple**.
 - **Starting Level:** The robot starts the session with 60% battery.
 - **Recharging:** The robot can recharge at any time. See **Recharging Protocol** for more details.
 - **Battery Consumption:** Interacting with large bricks reduces the battery by 40%, while small bricks reduces the battery by 30%. However, actions such as mapping, searching, or returning to the dock without a brick do not consume battery.
 - **Depleted Battery:** The team loses points if the robot's battery depletes (see rubric).
- **Moving Goal Penalty:** Moving a goal results in a penalty (see rubric).
- **Humans Moving Bricks:** You can expect a referee to move:
 - Any brick in the loading dock, even if the robot is approaching that brick.
 - Any brick from the goal zone back to the appropriate loading dock. You will not lose points if a brick is placed back into the loading dock.

4. Recharging Protocol

The robot must return to the **Recharge Station** (marked by a specific AprilTag) before its battery depletes. To ensure a successful charge:

1. The robot must approach the Recharge Tag **head-on**.

2. The head-on approach must begin from at least **30 centimeters away** from the tag.
3. The robot must be within **five (5) centimeters** of the tag and remain **stationary** for **5 secs.**
4. Once completed, the battery is restored to **100%.**

5. Requirements & Grading

The robot must be autonomous! You may reset it if it gets stuck, but no live coding is allowed.

Rubric

Criteria	Points
Mapping Environment: Map the following four types of items: two goals (4 pts each), recharge station (5 pts), loading dock (4 pts), and two obstacles (fabric box obstacles) (4 pts each).	25 pts
Battery Level Management: You automatically have these points. However, if your battery ever depletes (reaches 0%), you will automatically lose these points, will not be able to regain these points, and will not receive points for transporting bricks after the battery depletes. If the robot does not follow the recharging protocol as described, you'll only earn 3 pts.	15 pts
Task Efficiency: Deliver 5 bricks (2 pts each).	10 pts
Collision-Free: The robot does not: cross the perimeter (5 pts), touch any obstacle fabric box (5 pts), or touch any goals (5 pts).	15 pts

Competition/Live Demo: Demonstrate that your robot can deliver at two bricks: 1) Complete delivering at least two bricks (4 pts total). 2) Complete the demo without depleting the battery (5 pts). 3) Complete the demo without touching the boundary (2 pts), obstacles (2 pts), or goals (2 pts). 4) Generate a code-based map using the robot's observations (5 pts).	20 pts
Report	15 pts

6. Submission Guidelines

Teams must submit a video demonstration and a PDF report. The report must include:

- **Block Diagram** of the system architecture showing how the system works (from the camera to the decision-making processes).
- **Bird's Eye View Map** of the environment, containing the two obstacles (fabric boxes), goals (delivery zones), loading dock, and recharging station. You must include:
 - A photo of the environment, and
 - A map generated by code (e.g., via matplotlib in Python).
 - Only include the mapped items below (no trajectory is needed).
 - Have xy grid lines in spaces of 10 centimeters.
 - The center position of the shapes below should correspond to the center position of the object in the real world.
 - Objects include:
 - xy-axes representing the boundary of the environment,
 - Red circles for obstacles inside the environment,
 - Blue triangle for small goal,
 - Green triangle for large goal,
 - Yellow square for loading dock, and
 - Black square for recharging station.
- **Methodology** explaining the vision pipeline (YOLO/Segmentation), the battery state machine, and mapping strategy. Your pipeline components should reference the different blocks in your block diagram from above. The mapping strategy must describe how you used the measurements from the robot to directly create and maintain your map and include the corresponding code.

- **Results** including the total score, time, resets, and battery efficiency.
- **Appendix:** Your full code.

Collaboration Policy: You may discuss strategies with other teams, but all code and reports must be original to your team. Plagiarism checkers will be used.

7. Competition Guidelines and Scoring

Each group will be ranked based on the number of points you scored and the time it took for your robot to complete the task. The team with the **highest score** within the **least amount of time** and **resets** wins the competition. Each team has **five (5) minutes** to complete the task, which includes resets.

Competition Scoring	Points
Delivering Bricks: Success deliveries for: 1) Small bricks = 4 pts 2) Large bricks = 7 pts	4 or 7 pts
Battery Level Management: Same as rubric above.	15 pts
Collisions: The robot crosses the perimeter, touches any obstacle, or touches any goals.	-3 pts each time
Resets: Do not affect your score. However, remember that you are time-limited!	n/a